# ZERO-TOUCH ONBOARDING (ZTO)

## SECURE DEVICE ONBOARDING WITH mTLS & WEBSOCKETS

*Internship Experience and Learning Outcomes*


Nikhil Sharma
Roll No. 102203474

incedo

Computer Engineering Department
Thapar Institute of Engineering & Technology, Patiala
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mTLS
WebSocket
Zero-Touch

## ABOUT THE ORGANIZATION

incedo

Incedo is a global AI, data science, and technology services company that partners with enterprises to drive digital transformation and deliver measurable business outcomes.

**Digital Transformation Partner**
Helps organizations connect complex technology with real business goals.

**AI & Data Platforms**
Builds intelligent platforms and solutions that enable smarter automation and data-driven decisions.

**Technology Modernization**
Modernizes legacy workflows and systems to deliver secure, scalable and efficient operations.

**Operational Excellence**
Solves critical business challenges to improve efficiency and reduce operational costs.

**Industry Expertise**
Delivers specialized solutions for Banking, Life Sciences, Telecom and Hi-Tech industries.

## ROLES & RESPONSIBILITIES

- Built and maintained backend services using Golang for Zero-Touch Onboarding.
- Implemented secure communication using mTLS with certificate-based authentication.
- Designed real-time bi-directional communication using WebSockets.
- Worked on device lifecycle management, certificate provisioning using Trustpoint and validation.
- Implemented structured logging and monitoring using runtime logs.
- Collaborated in Agile ceremonies and tracked tasks using Jira.
- Used Git for version control and collaborative development.

## WHAT I DID

**KEY ACTIVITIES**

**Backend Development (Golang)**
Built REST APIs and microservices for ZTO solution.

**Secure Communication (mTLS)**
Implemented mTLS for authenticated and encrypted communication.

**Real-time Communication (WebSockets)**
Built WebSocket-based bi-directional channel for device updates.

**Certificate Management (Trustpoint & OpenSSL)**
Managed device certificates and automation using Trustpoint and OpenSSL.

## TECH STACK & TOOLS

| | |
|---|---|
| Golang | Backend development using Golang for scalable services |
| mTLS | Mutual TLS authentication for secure device onboarding |
| WebSockets (WSS) | Real-time bi-directional communication channel |
| OpenSSL | Certificate generation and PKI hierarchy creation |
| Trustpoint | Certificate provisioning and lifecycle management |
| Docker | Containerized services for consistent deployments |
| Git | Version control and collaborative development |
| Jira | Sprint planning and task management |
| Runtime Logs | Debugging, validation, and onboarding analysis |

## WHAT I LEARNED

**Backend Development**
Built scalable services and REST APIs using Golang.

**Secure Communication**
Implemented mTLS for secure and authenticated communication.

**Real-time Systems**
Gained hands-on experience with WebSockets for real-time updates.

**Certificate Management**
Understood certificate lifecycle using Trustpoint and OpenSSL.

**DevOps Basics**
Containerized applications using Docker for consistent deployments.

**Tools & Practices**
Used Git, Jira and runtime logs for effective collaboration, tracking and debugging.

## KEY OUTCOMES

- Built a secure and reliable ZTO solution using mTLS and WebSocket communication.
- Improved onboarding efficiency with automated certificate provisioning using Trustpoint.
- Enhanced system reliability with structured logging and monitoring using runtime logs.
- Strengthened understanding of secure communication, backend systems and certificate lifecycle management.
- Actively contributed in Agile ceremonies and delivered iterative improvements.

## KEY ACHIEVEMENTS

- Successfully implemented Zero-Touch Onboarding using mTLS and WebSocket.
- Automated certificate provisioning and lifecycle management using Trustpoint.
- Contributed to backend features, issue resolution and system validation.
- Enhanced debugging efficiency using structured runtime logging and analysis.
- Actively collaborated in Agile sprints and on-time delivery of tasks.

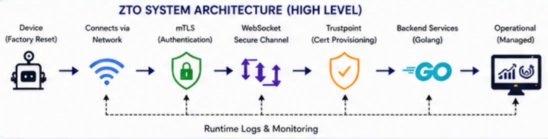
## WHAT COULD BE IMPROVED

- Explore advanced scaling strategies for high device concurrency.
- Implement advanced monitoring and alerting for proactive issue detection.
- Improve automation in certificate renewal and revocation.
- Expand test coverage and add more integration & stress tests.

## WHAT ADDITIONAL KNOWLEDGE HELPED

- Golang fundamentals and concurrency model
- Secure communication protocols (mTLS, TLS)
- WebSocket protocol and real-time systems
- PKI, certificates, and cryptography basics
- DevOps tools and containerization with Docker
- Agile methodology and version control best practices

## ZTO SYSTEM ARCHITECTURE (HIGH LEVEL)




## KEY TAKEAWAY

This internship strengthened my skills in secure backend development, real-time communication with WebSockets, and certificate management using mTLS and Trustpoint. It enhanced my problem solving, debugging and collaboration skills in a professional environment.


REFERENCES
[1] Go Documentation – https://golang.org/doc/
[2] WebSocket Protocol (RFC 6455) – https://datatracker.ietf.org/doc/html/rfc6455
[3] mTLS Overview – https://www.cloudfare.com/learning/ssl/what-is-mtls/
[4] OpenSSL Documentation – https://www.openssl.org/docs/
[5] Trustpoint – PKI & Certificate Lifecycle Management – https://www.trustpoint.com/


*"The best way to learn is by doing, and the best place to do it is where real problems create real impact."*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